

2 SETTING UP AND GETTING STARTED WITH PYTHON

This chapter shows the required steps to make your system ready and to write Python scripts. Sections of this chapter are segregated as follows:

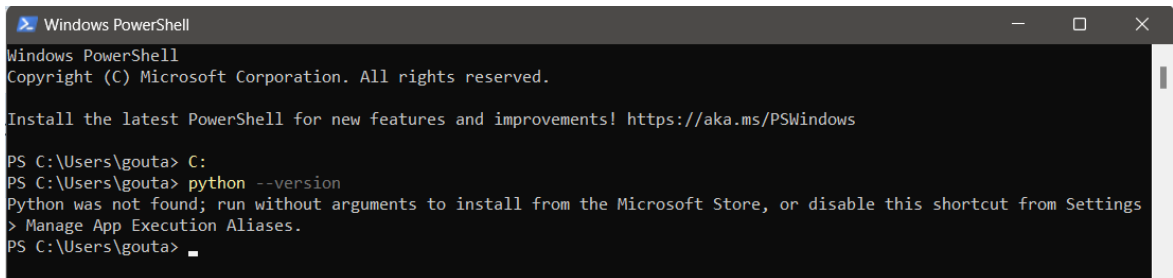
- First check your system comes with any version of Python
- Install or update Python on Windows, macOS, and Linux
- Use Python on mobile devices like phones or tablets
- Use Python on the Web with online interpreters

2.1 Installing Python on Windows

Before start to install, it is better to check if you already have Python on your Windows machine, first open a command-line application, such as PowerShell.

- Press the Windows  key
- Type PowerShell
- Press Enter Key 

Instead of PowerShell alternately you can also use cmd.exe or windows Terminal. With the command line open, type in the following command and press Enter :



```
Windows PowerShell
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PS C:\Users\gouta> C:
PS C:\Users\gouta> python --version
Python was not found; run without arguments to install from the Microsoft Store, or disable this shortcut from Settings
> Manage App Execution Aliases.
PS C:\Users\gouta> _
```

It seems Python is not available in your system, and you can find three ways to install the official Python distribution on Windows:

- Microsoft Store package: The most straightforward installation method on Windows involves

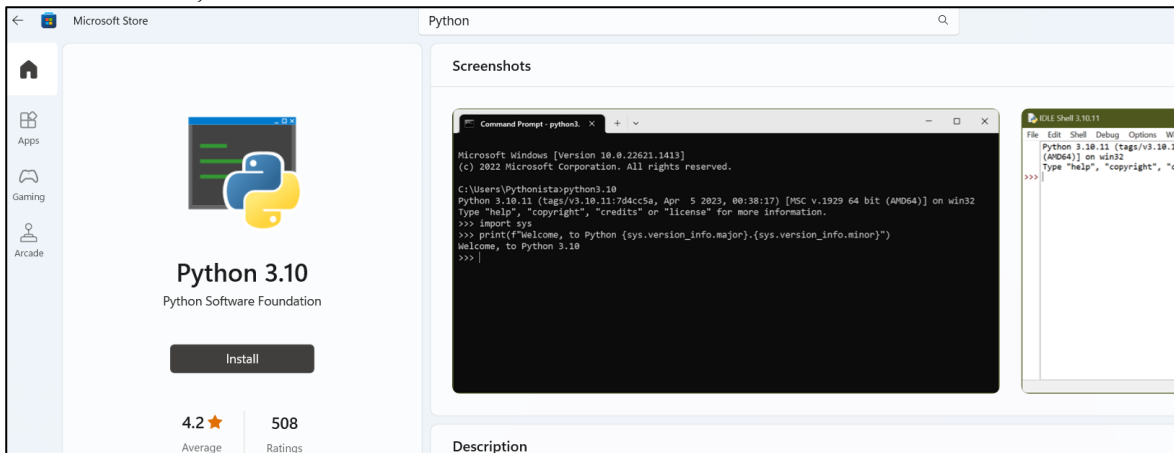
installing from the Microsoft Store app. This is recommended for beginner Python users looking for an easy-to-set-up interactive experience.

- Full Installer: This approach involves downloading Python directly from the Python.org website. This is recommended for intermediate and advanced developers who need more control during the setup process.

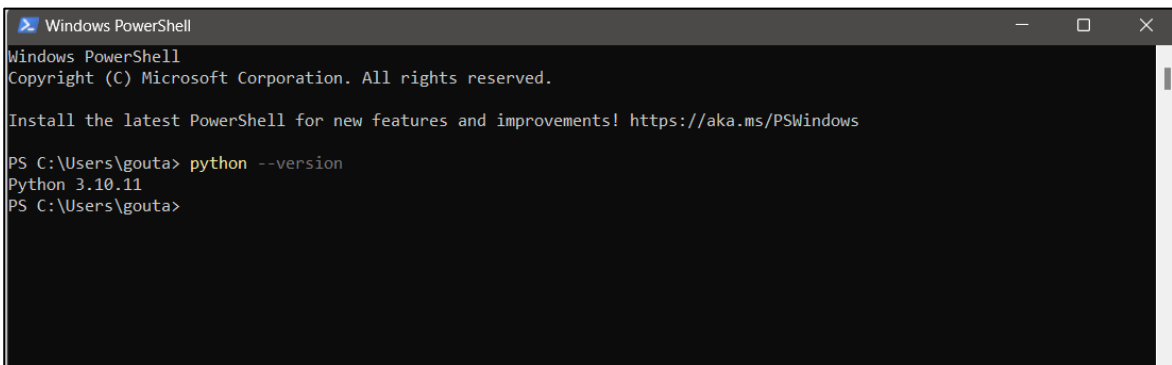
2.1.1 Installing from Microsoft App Store

If you're new to Python and looking to get started quickly, then the Microsoft Store package is the best way to get up and running without any fuss. You can install from the Microsoft Store in two steps.

- Open the Microsoft App Store, click  from Taskbar or alternately type python in PowerShell and press Enter ↵. The Microsoft Store will automatically launch and take you to the latest version of Python in the store. Select the most latest version that you can see or at least Python ver. 3.10, as all illustrations are tested on this version.



- Install the Python App and to do that click over **Install**. You can see downloading is start and wait for it. Make sure that the Python application you've selected is created by the **Python Software Foundation**.
- To check Python is installed or not type the given command **python --version** in the PowerShell. You can see the installed version, if installation is completed.



2.1.2 Installing from Full Installer

Follow the following steps:

- Open a browser window and navigate to the Python.org Downloads page for Windows.
- Under the **Download Tab** select **Windows** and till the writing of this chapter you can see latest stable version is 3.12.2. If you want any specific version, check the right side of the page as **Pre-releases heading**.
- Now select either Windows Installer 64-bit executable installer for 64-bit or Windows Installer 32-bit for 32-bit processor architecture.

You can also complete the installation on Windows using alternative distributions, such as Anaconda, but this chapter covers only official distributions. Anaconda¹⁹ is a popular platform for doing scientific computing and data science with Python. To download you can visit anaconda download page <https://www.anaconda.com/download>.

2.2 Installing Python in Ubuntu

In this section, you'll learn how to install Python using Ubuntu's apt package manager. Depending on the version of the Ubuntu distribution you run, the process for setting up Python on your system will vary. You can determine your local Ubuntu version by running the following command:

```
$ lsb_release -a
```

No LSB modules are available.

Distributor ID: Ubuntu

Description: Ubuntu 16.04.4 LTS

Release: 16.04

Codename: xenial

Follow the instructions below that match the version number you see under Release in the console output. Ubuntu 18.04, Ubuntu 20.04 and above, doesn't come by default Python 3.8, but it is available in the Universe repository. To install version 3.8, open a terminal application and type the following commands:

```
$ sudo apt-get update
```

```
$ sudo apt-get install python3.8 python3-pip
```

Once the installation is complete, you can run Python 3.8 with the python3.8 command and pip with the pip3 command.

2.3 Installing Python on macOS

Like Windows, in macOS also you can install it from direct installer by navigate to the Python.org Downloads page for macOS. Select any specific version under Pre-releases heading and download and install it. Before that, it is better to check any existing installer in available or not. To check which Python version you have on your Mac, first open a command-line application, such as Terminal. To open the

¹⁹ <https://www.anaconda.com/>

Terminal, follow the steps given below:

- Press the **⌘** + **space** keys.
- Type Terminal
- Press Enter **↵** key.

Alternatively, you can open Finder and navigate to Applications → Utilities → Terminal. With the command line open, type in the following commands:

```
# Check the system Python version
$ python --version
# Check the Python 2 version
$ python2 --version
# Check the Python 3 version
$ python3 --version
```

If you have Python on your system, then one or more of these commands should respond with a version number.

2.4 Online Python Interpreters

If you want to try out the examples in this tutorial without setting up Python on your machine, then there are several websites that offer an online Python interpreter:

- Python.org Online Console
- Repl.it
- Python Fiddle
- Trinket
- Python Anywhere

These cloud-based Python interpreters may not be able to execute some of the more complex examples in this tutorial, but they're adequate for running most of the code and may be a nice way to get started. More information on using these sites is presented in the next tutorial in this series.

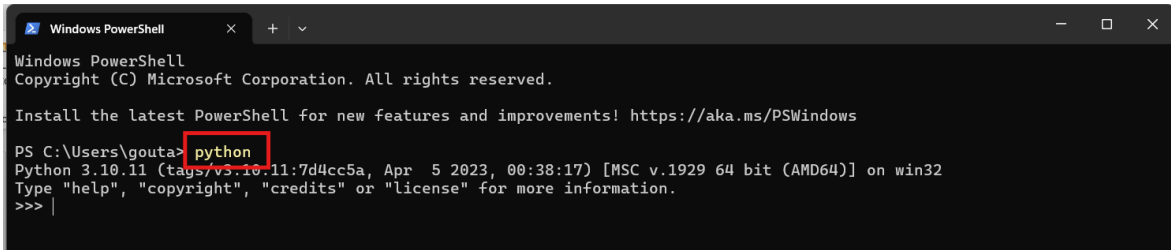
2.5 Executing Python Scripts

There are many ways you can execute a python script; these are as follows:

2.5.1 Executing Python Command using command line

Type PowerShell in windows search bar and open it as Administrator Mode

Then type python

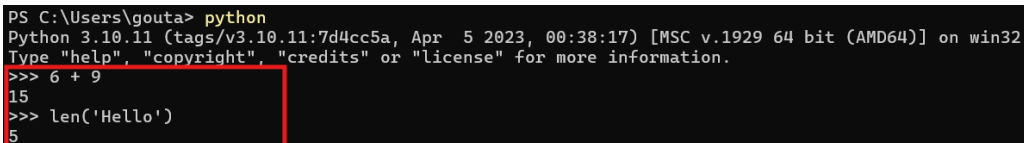


```
Windows PowerShell
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PS C:\Users\gouta> python
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr  5 2023, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

Now you can type any python command in the command prompt (>>>) and check the correct outcome, as shown below:



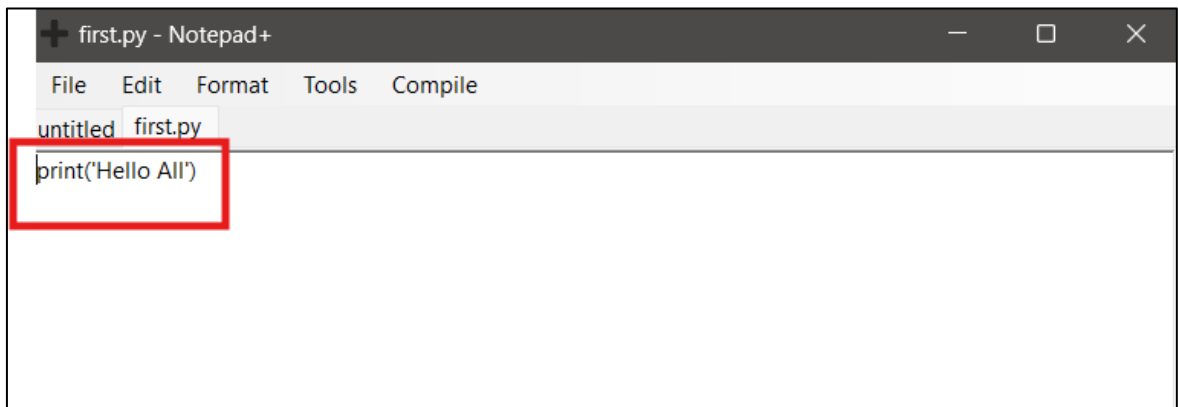
```
PS C:\Users\gouta> python
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr  5 2023, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> 6 + 9
15
>>> len('Hello')
5
```

2.5.2 Executing Python Script file using Command Line

You can use any text editor for writing your script and save the file with **.py** extension.

Let's assume that we have save the file a **first.py** in inside a folder **Python code** of **D:** drive.

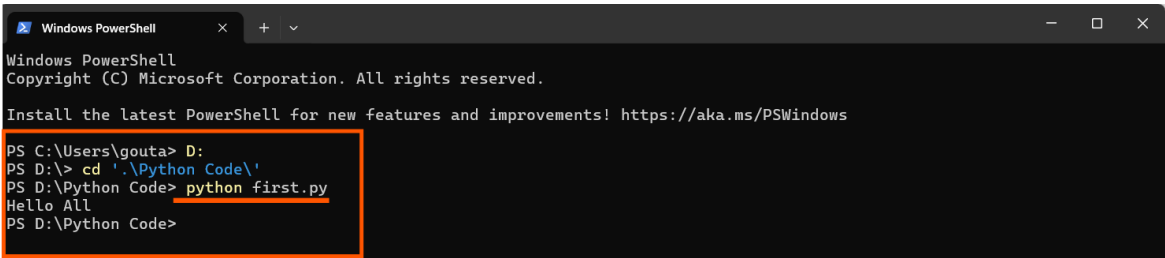
Now open any code editor such as Notepad++ and type python script and save it in location as **D:\Python code**



```
first.py - Notepad+
File Edit Format Tools Compile
untitled first.py
print('Hello All')
```

Now open PowerShell and set the path of the PowerShell to the desired location.

Now type the command as **python filename.py** in our case **python first.py**



```
Windows PowerShell
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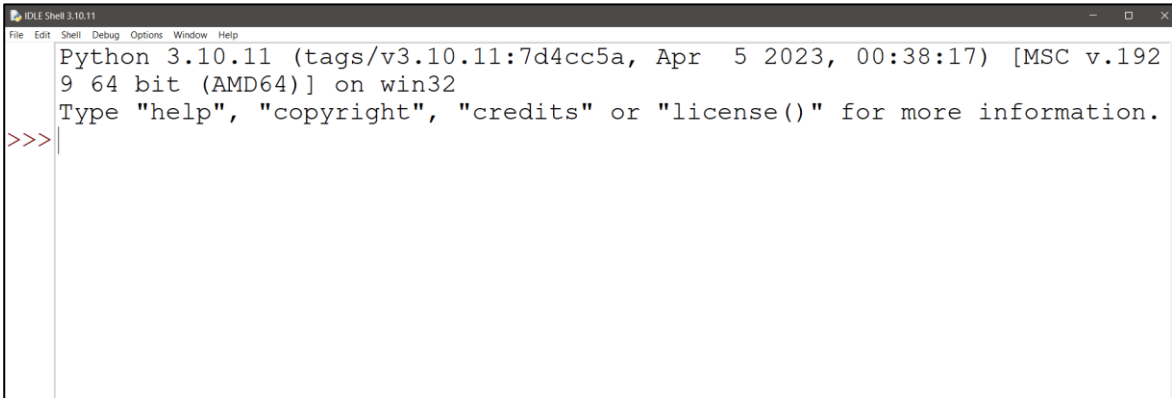
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\gouta> D:
PS D:\> cd '.\Python Code\'
PS D:\Python Code> python first.py
Hello All
PS D:\Python Code>
```

2.5.3 Executing Python Scripts using IDLE

If you follow the guide of python installation as discussed in Section 2.1 you can use Python Integrated Development and Learning Environment (IDLE) to write and execute any python script. Follow the steps given below:

Open the IDLE by clicking over the IDLE icon or search using windows search bar. Once it is open it will be look like this:



```
IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.192
9 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
```

You can directly type any python command after the Python command line (>>>), as shown below:



```
>>> a = 8
>>> b = 9
>>> a+b
17
>>>
```

If you are learning any programming language for the first time, it is better to write python command through script file. To open a new script file, you can open it by clicking over **File** menu -> **New File**.

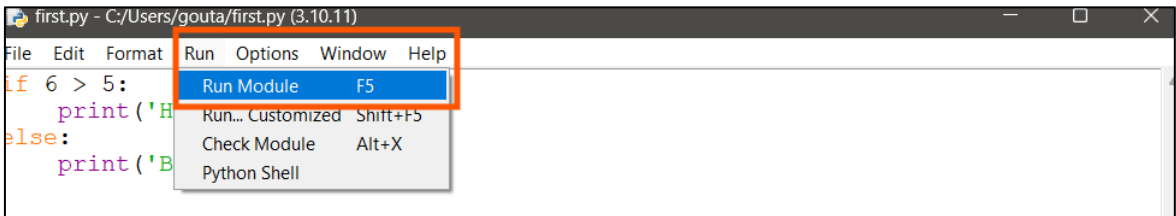


A screenshot of a Python script editor window titled 'first.py - C:/Users/gouta/first.py (3.10.11)'. The menu bar includes File, Edit, Format, Run, Options, Window, and Help. The script content is as follows:

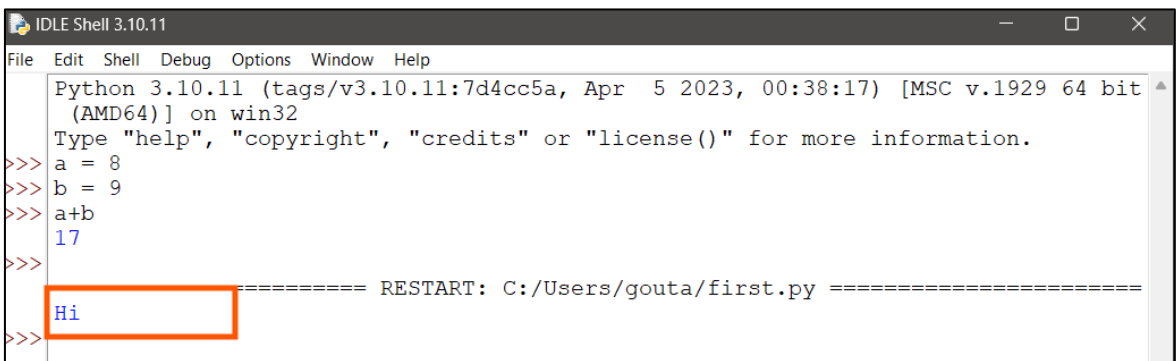
```
if 6 > 5:
    print('Hi')
else:
    print('Bye Bye')
```

Save the file and save it in your preferred location.

To execute the script Click **Run**→ **Run Module** or **Press F5** and output will display in the IDLE window.



A screenshot of the same Python script editor window with the 'Run' menu open. The 'Run Module' option, which is associated with the F5 key, is highlighted. Other options in the menu include 'Run... Customized' (Shift+F5), 'Check Module' (Alt+X), and 'Python Shell'.



A screenshot of the IDLE Shell window titled 'IDLE Shell 3.10.11'. The menu bar includes File, Edit, Shell, Debug, Options, Window, and Help. The shell displays the following output:

```
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> a = 8
>>> b = 9
>>> a+b
17
>>>
===== RESTART: C:/Users/gouta/first.py =====
>>> Hi
>>>
```

*****END OF CHAPTER*****